

Dr.-Ing. Ali Khalifa

Computational multiphase flows | Particle processes | Machine learning

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Research profile

Mechanical engineer and postdoctoral researcher working on predictive modelling of particulate flows. Research combines particle-resolved simulation, Euler–Lagrange modelling and machine learning to understand and model cohesive agglomerate breakage and collisions in turbulent flows. DFG Co-Principal Investigator with university lecturing and student supervision experience.

Academic appointments

Postdoctoral Researcher, Technical University of Munich Chair of Process Systems Engineering, Freising, Germany.	11.2025–present
Postdoctoral Researcher, Helmut-Schmidt University Department of Fluid Mechanics, Hamburg, Germany.	10.2022–10.2025
Doctoral Researcher, Helmut-Schmidt University Department of Fluid Mechanics, Hamburg, Germany.	12.2016–09.2022
Student Research Assistant, Otto von Guericke University Laboratory of Fluid Dynamics and Technical Flows, Magdeburg, Germany.	12.2015–10.2016

Competitive research funding

DFG Research Grant Co-Principal Investigator <i>Deagglomeration in turbulent flows: Particle-resolved investigation for detailed insights and advanced modeling.</i> German Research Foundation (DFG); reference KH 653/3-1; project 573724408.	12.2025–11.2028
▪ Secured funding for a student research assistant (HiWi) position at TUM and supervise the funded assistant. ▪ Contribute to supervision of the project's doctoral researcher at Helmut-Schmidt University, Hamburg.	
Internal Research Funding (IFF), Helmut-Schmidt University Competitively awarded one-year seed funding to develop an external research grant proposal; selected through a written application and oral presentation.	07.2023–06.2024

Education

Dr.-Ing., Helmut-Schmidt University, Hamburg With distinction; oral examination: 19.09.2022. Supervisor: Prof. Michael Breuer. Thesis: <i>Modeling and simulation of the breakage of cohesive particle agglomerates in turbulent wall-bounded flows.</i> DOI: 10.24405/14518 .	Sep. 2022
M.Sc., Chemical and Energy Engineering Otto von Guericke University, Magdeburg, Germany.	10.2013–04.2016
B.Sc., Mechanical Engineering University of Jordan, Amman, Jordan.	09.2008–01.2013

Teaching experience

Lecturer | Technical University of Munich

2026

Control Engineering (Regelungstechnik), summer semester 2026.

Machine Learning in Food and Life Sciences, winter semester 2026/27 (scheduled).

Teaching Assistant | Helmut-Schmidt University, Hamburg

Fluid Mechanics; Applied Fluid Mechanics.

Research supervision

Technical University of Munich

Alexander Todorov | Master's thesis

2026

Simulation of the atomization process for melt coating using OpenFOAM.

Julius Eckert | Student research assistant

2026

Algorithm for generating numerical agglomerates with specified fractal structures for DEM simulations.

Helmut-Schmidt University, Hamburg

Doctoral researcher | DFG project

Current

Contribute to doctoral supervision within the DFG-funded deagglomeration project in collaboration with the Hamburg team.

Hoang Huy Nguyen | Master's thesis

2025

Investigations on models for the breakup of particle agglomerates in homogeneous isotropic turbulence in the context of Euler–Lagrange simulations.

Hoang Huy Nguyen | Pre-master's research project

2025

Direct numerical simulation of homogeneous isotropic turbulence in a periodic box.

Jasper Gollwitzer | Master's thesis

2021

Application of artificial neural network-based models for agglomerate breakage by wall impact in Euler–Lagrange large-eddy simulations.

Jasper Gollwitzer | Pre-master's research project

2021

DEM simulations of agglomerate breakage by wall impact.

Constance Le Tacon | Pre-master's research project

2020

CFD simulations of fluid flow in configurations relevant to dry powder inhalers.

Honours and awards

Wissenschaftspreis (Science Award), Helmut-Schmidt University

2024

Co-recipient of the award for the best doctoral dissertation; awarded by Freunde und Förderer der Helmut-Schmidt-Universität / Universität der Bundeswehr e.V., March 2024.

Award for Social Excellence, University of Jordan

2013

Awarded by the University President for involvement in student organisations and extracurricular activities.

Leadership and service

Elected President, Student Union

2011–2012

University of Jordan, Amman.

Elected President, Energy Club

2010–2011

University of Jordan, Amman.

Co-founder and Manager, Mech-Family student organisation

2009–2010

University of Jordan, Amman.

Peer-reviewed journal articles

10 articles; 8 as first author

1. **A. Khalifa**, M. Breuer and H.H. Nguyen, 2026. Assessment of Euler–Lagrange models for fluid-induced cohesive particle deagglomeration using particle-resolved DNS. *Particuology* 114, 408–430. DOI: [10.1016/j.partic.2026.04.022](https://doi.org/10.1016/j.partic.2026.04.022)
2. **A. Khalifa** and M. Breuer, 2025. Investigation of cohesive particle deagglomeration in homogeneous isotropic turbulence using particle-resolved direct numerical simulation. *Physics of Fluids* 37, 083339. DOI: [10.1063/5.0278786](https://doi.org/10.1063/5.0278786)
3. **A. Khalifa** and M. Breuer, 2024. Evaluation of an efficient data-driven ANN model to predict agglomerate collisions within Euler–Lagrange simulations. *Computers & Fluids* 269, 106119. DOI: [10.1016/j.compfluid.2023.106119](https://doi.org/10.1016/j.compfluid.2023.106119)
4. **A. Khalifa** and M. Breuer, 2023. Data-driven ANN approach for binary agglomerate collisions including breakage and agglomeration. *Chemical Engineering Research and Design* 195, 14–27. DOI: [10.1016/j.cherd.2023.05.040](https://doi.org/10.1016/j.cherd.2023.05.040)
5. **A. Khalifa**, M. Breuer and J. Gollwitzer, 2022. Neural-network based approach for modeling wall–impact breakage of agglomerates in particle–laden flows applied in Euler–Lagrange LES. *Int. J. Heat Fluid Flow* 94, 108897. DOI: [10.1016/j.ijheatfluidflow.2021.108897](https://doi.org/10.1016/j.ijheatfluidflow.2021.108897)
6. **A. Khalifa**, J. Gollwitzer and M. Breuer, 2021. LES of particle-laden flow in sharp pipe bends with data-driven predictions of agglomerate breakage by wall impacts. *Fluids* 6 (12), 424. DOI: [10.3390/fluids6120424](https://doi.org/10.3390/fluids6120424)
7. **A. Khalifa** and M. Breuer, 2021. An efficient model for the breakage of agglomerates by wall impact applied to Euler–Lagrange LES predictions. *Int. J. Multiphase Flow* 142, 103625. DOI: [10.1016/j.ijmultiphaseflow.2021.103625](https://doi.org/10.1016/j.ijmultiphaseflow.2021.103625)
8. **A. Khalifa** and M. Breuer, 2020. Data-driven model for the breakage of dry monodisperse agglomerates by wall impact applicable for multiphase flow simulations. *Powder Technology* 376, 241–253. DOI: [10.1016/j.powtec.2020.07.089](https://doi.org/10.1016/j.powtec.2020.07.089)
9. M. Breuer and **A. Khalifa**, 2019b. Refinement of breakup models for compact powder agglomerates exposed to turbulent flows considering relevant time scales. *Computers & Fluids* 194, 104315. DOI: [10.1016/j.compfluid.2019.104315](https://doi.org/10.1016/j.compfluid.2019.104315)
10. M. Breuer and **A. Khalifa**, 2019a. Revisiting and improving models for the breakup of compact dry powder agglomerates in turbulent flows within Eulerian–Lagrangian simulations. *Powder Technology* 348, 105–125. DOI: [10.1016/j.powtec.2019.03.009](https://doi.org/10.1016/j.powtec.2019.03.009)

Peer-reviewed conference proceedings

1. **A. Khalifa** and M. Breuer, 2026. Efficient Euler–Lagrange LES of Agglomerate-Laden Flow in a T-Junction Including an ANN-Based Collisional Breakage. In: 14th Int. ERCOFTAC Workshop on DNS and LES: DLES-14, Erlangen, Germany, April 10–12, 2024, ERCOFTAC Series, Vol. 21. pp. 127, Direct and Large–Eddy Simulation XIV, eds. M.V. Salvetti, S. Hickel, and C. Marchioli, ISBN 978-3-032-19267-7, Springer Nature Switzerland AG, (2026).
2. **A. Khalifa** and M. Breuer, 2024. Euler–Lagrange LES predictions of a powder disperser including a multiscale wall-impact breakage model. In: 13th Int. ERCOFTAC Workshop on DNS and LES: DLES-13, Udine, Italy, October 26–29, 2022, ERCOFTAC Series, Vol. 31. pp. 89–94, Direct and Large–Eddy Simulation XIII, eds. C. Marchioli et al., ISBN 978–3–031–47027–1, Springer Nature Switzerland AG, (2024). DOI: [10.1007/978-3-031-47028-8_14](https://doi.org/10.1007/978-3-031-47028-8_14)
3. M. Breuer and **A. Khalifa**, 2020. Breakup of agglomerates in turbulent flows: An Euler–Lagrange LES study. Twelfth Int. ERCOFTAC Workshop on DNS and LES: DLES-12, Madrid, Spain, June 5–7, 2019, ERCOFTAC Series, vol. 27, pp. 127–133, Direct and Large–Eddy Simulation XII, eds. M. García-Villalba et al., ISBN 978–3–030–42821–1, Springer Nature Switzerland AG, (2020). DOI: [10.1007/978-3-030-42822-8_17](https://doi.org/10.1007/978-3-030-42822-8_17)